

Introductory Statistics for Econometrics

Unit 9. Confidence Intervals and Hypothesis Testing

CIs, hypothesis tests, p -values,
and inference on the OLS slope

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Learning outcomes. After this unit the student should be able to (i) construct and interpret a confidence interval; (ii) state the null and alternative hypotheses, identify a test statistic and its critical region; (iii) compute α , β , power and p -values; (iv) understand the duality between CIs and tests; (v) carry out a t -test and confidence interval for the OLS slope $\hat{\beta}_1$.

Source material. Condensed from MIT OCW 14.30 (Spring 2009), K. Menzel, Lecture Notes 19–23. The inference on $\hat{\beta}_1$ follows Stock and Watson, *Introduction to Econometrics*, Chapter 5.

1 Confidence intervals: definition

A point estimate $\hat{\theta}$ is one number. By itself, it conveys nothing about precision. A *confidence interval* packages the estimate with its standard error into a range that is likely to contain the true parameter.

Definition 1.1 (Confidence interval). A $1 - \alpha$ *confidence interval* for θ_0 is a random interval $[A(X), B(X)]$ depending on the sample with the property

$$\mathbb{P}_{\theta_0}(A(X) \leq \theta_0 \leq B(X)) = 1 - \alpha \quad \text{for every } \theta_0 \in \Theta.$$

The number $1 - \alpha$ is the *confidence level* (typically 0.90, 0.95, 0.99).

What a CI is, and what it isn't. A confidence interval is a way of measuring the uncertainty around an estimate. Suppose we want to estimate an unknown parameter θ_0 , and we use an estimator $\hat{\theta}$. Since $\hat{\theta}$ is computed from a random sample, different samples would generally produce different estimates.

The usual form of a confidence interval is

$$\hat{\theta} \pm \text{critical value} \times \widehat{\text{SE}}(\hat{\theta}).$$

The term

$$\text{critical value} \times \widehat{\text{SE}}(\hat{\theta})$$

is the margin of error. It measures how much sampling uncertainty we attach to the estimate. For example, in large samples we often use the normal approximation

$$\frac{\hat{\theta} - \theta_0}{\widehat{\text{SE}}(\hat{\theta})} \approx \mathcal{N}(0, 1).$$

Therefore, an approximate $(1 - \alpha)$ confidence interval is

$$\text{CI}_{1-\alpha} = \left[\hat{\theta} - z_{1-\alpha/2} \widehat{\text{SE}}(\hat{\theta}), \hat{\theta} + z_{1-\alpha/2} \widehat{\text{SE}}(\hat{\theta}) \right].$$

For a 95% confidence interval, $z_{0.975} \approx 1.96$, so we get the familiar expression

$$CI_{95\%} = \hat{\theta} \pm 1.96 \widehat{SE}(\hat{\theta}).$$

The interpretation of a confidence interval is important. A 95% confidence interval does not mean that there is a 95% probability that θ_0 lies inside the particular interval we computed. After the sample is observed, the interval is fixed and the parameter θ_0 is also fixed. Instead, the interpretation is a repeated-sampling one. If we repeatedly drew samples from the population and constructed a confidence interval from each sample, then approximately 95% of those intervals would contain the true parameter θ_0 .

When the sample is small and the standard error is estimated, we often replace the normal critical value by a t critical value:

$$CI_{1-\alpha} = \hat{\theta} \pm t_{n-1, 1-\alpha/2} \widehat{SE}(\hat{\theta}).$$

The t distribution has heavier tails than the normal distribution, so the resulting confidence interval is wider. As the sample size increases, the t critical value becomes very close to the normal critical value. In econometrics, the main practical difficulty is often not the choice between 1.96 and a t critical value. The more important question is whether the standard error is appropriate for the setting. Depending on the problem, we may need homoskedastic standard errors, heteroskedasticity-robust standard errors, clustered standard errors, or other corrections.

Example 1.2 (Polling margin of error). Suppose a poll surveys $n = 4,385$ voters. Let $\bar{X}_n$ denote the sample proportion supporting a candidate, and let p be the true population proportion.

For a binary variable,

$$\text{Var}(\bar{X}_n) = \frac{p(1-p)}{n}.$$

The largest possible value of $p(1-p)$ is $1/4$, which occurs when $p = 0.5$. Therefore,

$$SE(\bar{X}_n) \leq \sqrt{\frac{1}{4n}} = \sqrt{\frac{1}{4 \cdot 4,385}} \approx 0.0076.$$

Using the normal approximation, the half-width of a 95% confidence interval is at most

$$1.96 \times 0.0076 \approx 0.015.$$

Thus, the margin of error is about 1.5 percentage points.

Example 1.3 (Sample size needed for a given precision). Suppose a measurement device has known standard deviation

$$\sigma = 0.005.$$

We want a 90% confidence interval with half-width at most 0.0005.

For a 90% confidence interval, the normal critical value is approximately

$$z_{0.95} = 1.645.$$

The half-width of the confidence interval is therefore

$$1.645 \frac{\sigma}{\sqrt{n}} = 1.645 \frac{0.005}{\sqrt{n}}.$$

We require this to be no larger than 0.0005:

$$1.645 \frac{0.005}{\sqrt{n}} \leq 0.0005.$$

Solving for n gives

$$n \geq \left(\frac{1.645 \cdot 0.005}{0.0005} \right)^2 \approx 271.$$

Therefore, we need at least 271 readings.

2 Hypothesis testing: setup

A *statistical hypothesis test* is a decision rule: given the data, do we side with the null hypothesis H_0 or the alternative H_A ?

Definition 2.1 (Hypotheses, test statistic, critical region). A *null hypothesis* H_0 specifies a subset $\Theta_0 \subseteq \Theta$ the true parameter is conjectured to belong to; the *alternative* H_A specifies a disjoint subset Θ_A . A *test statistic* $T_n = T(X_1, \dots, X_n)$ summarises the sample. The *critical region* $C \subseteq \mathbb{R}$ is the set of values of T_n for which we *reject* H_0 . *Simple* hypothesis: Θ_0 is a single point. *Composite*: Θ_0 contains many points.

Definition 2.2 (Two types of error).

$$\alpha = \mathbb{P}_{H_0}(\text{reject } H_0) = \Pr(\text{Type I}), \quad \beta(\theta) = \mathbb{P}_{\theta \in \Theta_A}(\text{do not reject } H_0) = \Pr(\text{Type II} \mid \theta).$$

The *significance level* is α , the *power function* is $\theta \mapsto 1 - \beta(\theta)$.

Asymmetric treatment of the two errors. The convention is to control α at a small level (5% or 1%) and seek the test with the highest power. Type I error is treated as the “serious” error: a false rejection of an established claim. Think on Type I error as sending an innocent to jail while type II leaving a guilty free.

3 One-sided and two-sided tests

Example 3.1 (Two-sided z -test). $X_1, \dots, X_n$ i.i.d. $\mathcal{N}(\mu, \sigma^2)$ with σ^2 known. Test $H_0 : \mu = \mu_0$ against $H_A : \mu \neq \mu_0$. Use $Z = \sqrt{n}(\bar{X}_n - \mu_0)/\sigma$, which is $\mathcal{N}(0, 1)$ under H_0 . Reject when $|Z| > z_{1-\alpha/2}$. For $\alpha = 0.05$ the rule is $|Z| > 1.96$.

Example 3.2 (One-sided z -test). Same setup, but $H_A : \mu > \mu_0$. Reject when $Z > z_{1-\alpha}$. For $\alpha = 0.05$, $z_{0.95} = 1.645$. The test has higher power against $\mu > \mu_0$ than the two-sided test, at the cost of being uninformative against $\mu < \mu_0$.

Unknown variance: the t -test. Replace σ by the sample standard deviation $S_n = \sqrt{\frac{1}{n-1} \sum (X_i - \bar{X}_n)^2}$. The statistic $t = \sqrt{n}(\bar{X}_n - \mu_0)/S_n \sim t_{n-1}$ under H_0 (when X_i are normal). Critical values come from the t_{n-1} table; for $n \geq 30$ they are essentially identical to the normal table.

4 p -values

Definition 4.1 (p -value). The p -value is the smallest significance level at which the observed sample would lead to rejection of H_0 . Equivalently, the probability under H_0 of obtaining a value of the test statistic at least as extreme as the one observed.

For a two-sided z -test with observed value z^{obs} : $p = 2[1 - \Phi(|z^{\text{obs}}|)]$. The decision rule “reject if $p < \alpha$ ” is exactly equivalent to “reject if $|z^{\text{obs}}| > z_{1-\alpha/2}$ ”.

Remark 4.2 (Why p -values matter). Reporting only “reject at 5%” throws away information. The p -value quantifies the strength of evidence against H_0 : a $p = 0.049$ is borderline, a $p = 0.0001$ is overwhelming. Modern empirical practice reports p -values directly (or, equivalently, the test statistic and the user picks α).

5 Duality: CI *is* hypothesis test

Proposition 5.1 (Duality). *A $1 - \alpha$ confidence interval $\text{CI}_{1-\alpha}(X)$ for θ and a level- α two-sided test of $H_0 : \theta = \theta_0$ are equivalent:*

$$\theta_0 \notin \text{CI}_{1-\alpha}(X) \iff \text{reject } H_0 \text{ at level } \alpha.$$

This is a useful diagnostic: any 95% CI provides, for free, a 5% test against every value of θ_0 outside it. Conversely, the set of all θ_0 that the level- α test *fails to reject* is a $1 - \alpha$ confidence set.

6 Inference on the OLS slope (Stock–Watson Ch. 5)

We now apply the toolkit to the simple linear regression model $Y_i = \beta_0 + \beta_1 X_i + u_i$ from Unit 9 under the Least-Squares Assumptions (LSA1)–(LSA3).

Theorem 6.1 (Sampling distribution of $\hat{\beta}_1$). *Under (LSA1)–(LSA3) with $\text{Var}(X_i) > 0$, as $n \rightarrow \infty$,*

$$\frac{\hat{\beta}_1 - \beta_1}{\text{SE}(\hat{\beta}_1)} \xrightarrow{d} \mathcal{N}(0, 1),$$

where $\text{SE}(\hat{\beta}_1)$ is the asymptotic standard error of $\hat{\beta}_1$.

The estimated standard error $\widehat{\text{SE}}(\hat{\beta}_1)$ is the heteroskedasticity-robust formula from Unit 9.

Confidence interval for β_1 . For large samples,

$$\text{CI}_{1-\alpha}(\beta_1) = [\hat{\beta}_1 - z_{1-\alpha/2} \widehat{\text{SE}}(\hat{\beta}_1), \hat{\beta}_1 + z_{1-\alpha/2} \widehat{\text{SE}}(\hat{\beta}_1)].$$

At $\alpha = 0.05$: $\hat{\beta}_1 \pm 1.96 \widehat{\text{SE}}(\hat{\beta}_1)$. This is what Stata’s `regress` reports next to each coefficient.

Two-sided t -test of $H_0 : \beta_1 = \beta_{1,0}$.

$$t = \frac{\hat{\beta}_1 - \beta_{1,0}}{\widehat{\text{SE}}(\hat{\beta}_1)} \xrightarrow[H_0]{d} \mathcal{N}(0, 1).$$

Reject at level α if $|t| > z_{1-\alpha/2}$. p -value: $p = 2[1 - \Phi(|t|)]$. The default “significance test” in regression output is $H_0 : \beta_1 = 0$ vs. $H_A : \beta_1 \neq 0$, asking whether X helps predict Y at all.

Example 6.2 (Inference on the bus-commute regression). From Unit 9 (notes), regressing commute time Y on number of buses X with $n = 60$ days gave $\hat{\beta}_1 = 6.0$ minutes per bus. Suppose $\widehat{\text{SE}}(\hat{\beta}_1) = 1.5$. Then:

- 95% CI: $6 \pm 1.96 \times 1.5 = [3.06, 8.94]$ minutes per bus.
- Test of $H_0 : \beta_1 = 0$: $t = 6/1.5 = 4$, $p \approx 6 \times 10^{-5}$ — reject decisively. The number of buses is highly significant.
- Test of $H_0 : \beta_1 = 5$: $t = (6 - 5)/1.5 = 0.67$, $p \approx 0.50$ — *do not* reject. The data are consistent with each bus adding 5 minutes.

Joint hypotheses (F -test). With k regressors a hypothesis may involve several coefficients simultaneously, e.g. $H_0 : \beta_1 = \beta_2 = 0$. The F -statistic generalises t^2 :

$$F = (R\hat{\beta} - r)^\top [R\widehat{\text{Var}}(\hat{\beta})R^\top]^{-1} (R\hat{\beta} - r) / q,$$

which is asymptotically χ_q^2/q under H_0 (so the F distribution is the finite-sample version under normality). The default “ F -test of overall significance” reported by regression software is $H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$.

Summary

- A *confidence interval* packages estimate + standard error into a range that covers θ_0 with prescribed probability. Four standard cases: known σ (z), unknown σ (t), large-sample CLT (z with estimated SE), and exact (invert the c.d.f.).
- A *hypothesis test* controls Type I error at level α and seeks high power. p -value: smallest α at which the test rejects.
- One-sided vs two-sided: choose by what alternative is meaningful. In economics, the default is two-sided.
- *Duality*: $\theta_0 \in \text{CI}_{1-\alpha} \Leftrightarrow$ level- α test does not reject $H_0 : \theta = \theta_0$.
- *Neyman–Pearson lemma*: among tests of size α for simple-vs-simple hypotheses, the likelihood ratio test is most powerful. The generalised LR test extends to composite hypotheses.
- *OLS inference* (Stock–Watson Ch. 5): $\hat{\beta}_1$ is asymptotically normal under (LSA1–3). Build CIs and t -tests using the heteroskedasticity-robust standard error. Joint hypotheses use the F -statistic.

Suggested exercises

1. A sample of $n = 64$ from $\mathcal{N}(\mu, 16)$ has $\bar{X} = 102$. Build a 95% CI for μ . Test $H_0 : \mu = 100$ vs. $H_A : \mu \neq 100$.
2. In Example 1.2, what sample size guarantees a 95% margin of error of at most 1 percentage point?
3. Show that the two-sided z -test in Example 3.1 has power function $1 - \beta(\mu) = \Phi(\sqrt{n}(\mu - \mu_0)/\sigma - z_{1-\alpha/2}) + \Phi(-\sqrt{n}(\mu - \mu_0)/\sigma - z_{1-\alpha/2})$.
4. (Duality.) Verify Proposition 5.1 for the known-variance normal case by explicit algebra.
5. (*Stock–Watson Ch. 5 style.*) Using the same simulation as Unit 9 Exercise 6, simulate 1,000 samples of size $n = 200$ from $Y = 1 + 2X + u$ and verify that the empirical coverage of the 95% CI for β_1 is approximately 95%.
6. In Example 6.2, suppose a follow-up data set with $n = 200$ days gives $\hat{\beta}_1 = 6.5$ and $\widehat{\text{SE}}(\hat{\beta}_1) = 0.5$. Build the new 95% CI and test $H_0 : \beta_1 = 5$.